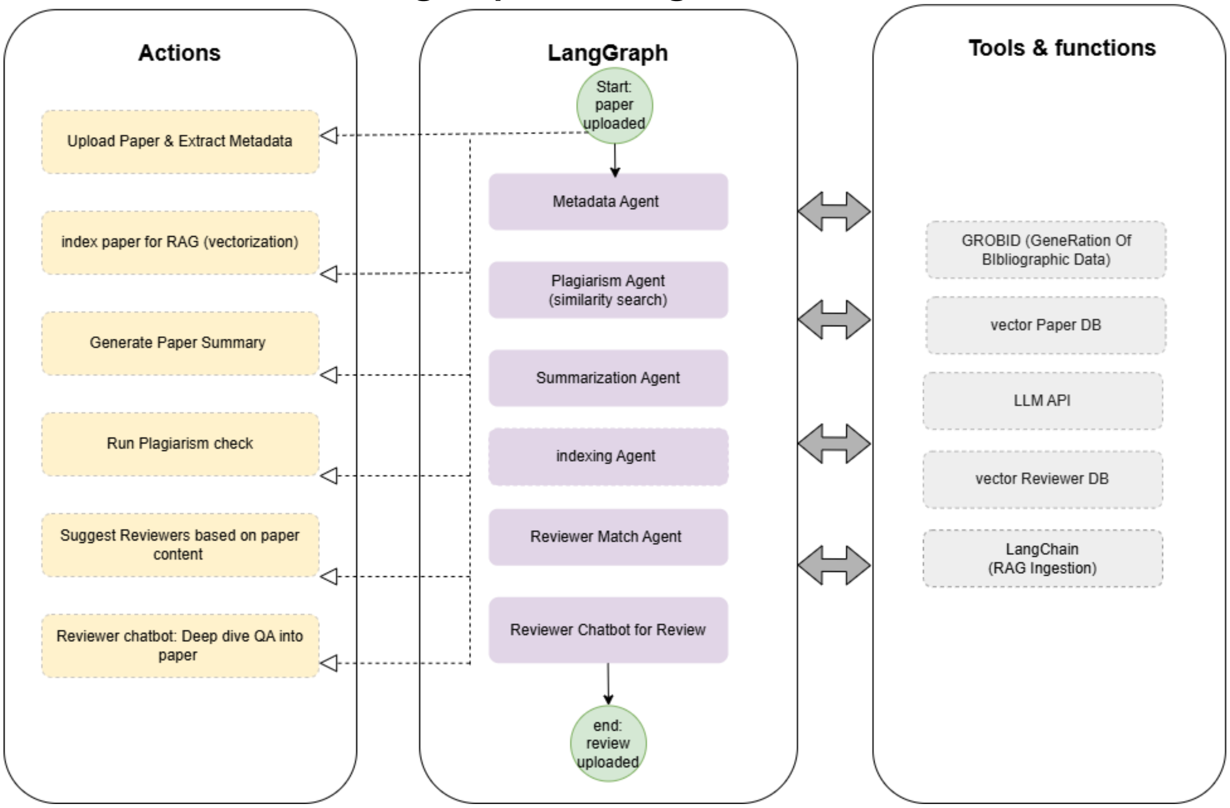
LangGraph Multi-Agent Workflow



Actions:

1. **Upload Paper & Extract Metadata**
   - User uploads a PDF; the system automatically extracts title, authors, abstract, and other bibliographic data using **GROBID**.

GROBID – A machine-learning tool for extracting metadata from scholarly PDFs.

2. **Index Paper for RAG (Vectorization)**
   - The paper's full text is chunked, embedded, and stored in a vector database to enable semantic search for later QA.
3. **Generate Paper Summary**
   - A large language model (LLM) produces a concise 2–3 paragraph summary of the paper's core contributions.
4. **Run Plagiarism Check**
   - The paper is compared against a local corpus using **embedding similarity**.
5. **Suggest Reviewers Based on Paper Content**
   - Using **RAG**, the paper's abstract/full text is matched against reviewer profiles (based on their expertise) to recommend the top 5 most relevant reviewers.
6. **Reviewer Chatbot: Deep Dive QA into Paper**
   - A chatbot allows reviewers to ask natural-language questions about the paper. The system retrieves relevant chunks from the paper's vector store and generates grounded answers via an LLM.

LangChain (RAG Ingestion) – Framework used to orchestrate the RAG pipeline: chunking, embedding, retrieval, and LLM integration.